Tunabot-A1 Turning Capabilities and Control

Dakota Chou, Joseph Zhu, Hilary Bart-Smith

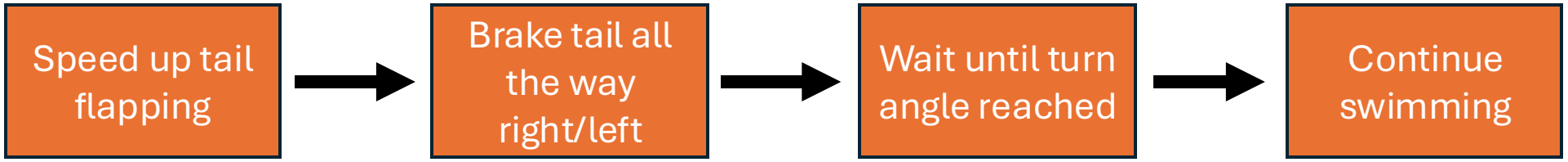
MURI Review, September 21, 2026

Objectives

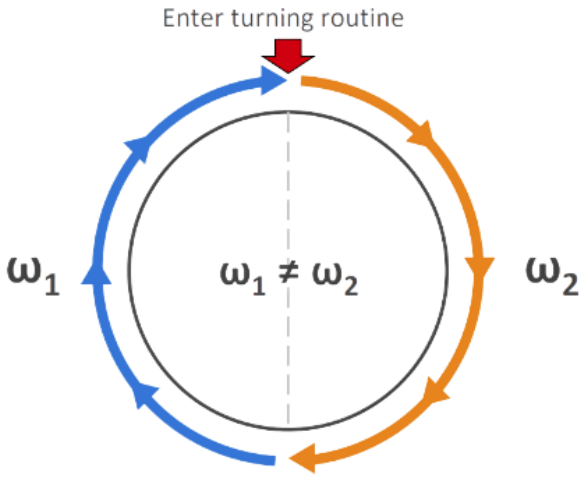
- Analyze current turning method to find improvements
- Evaluate success of improvements through experimentation
- Use improved capabilities for precise navigation

Current Turning Systems

- Burst and wait
  - Achieved turning radius of ~2 body lengths
  - Unable to achieve consistent turns



- Asymmetric flapping (variable speed during flap)
  - Achieved turning radius of 1 body length on a simpler version

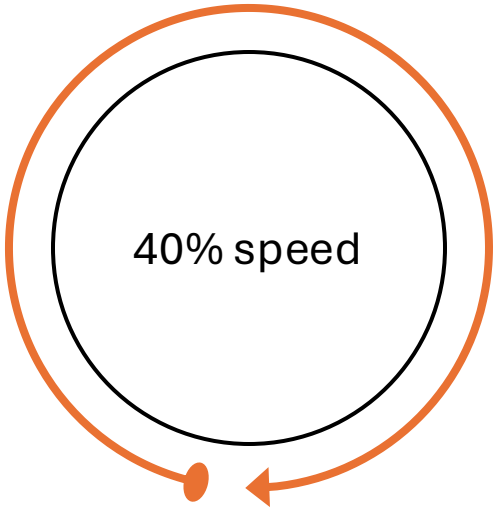


Development of a turning control strategy for a bio-inspired underwater vehicle O McKenney, J Zhu, T Han, H Bart-Smith Bioinspiration & Biomimetics 20 (5), 056004

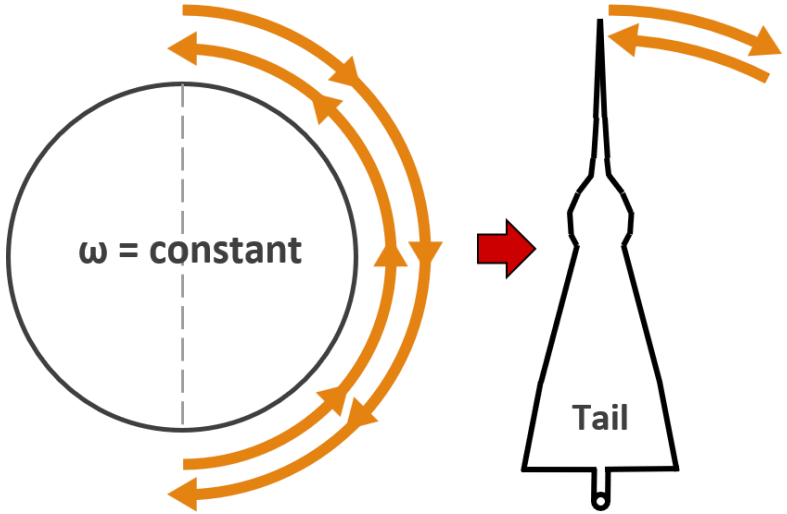
Proposed System

Half cycle flaps made possible by improved motor encoders and faster processing from the Arduino

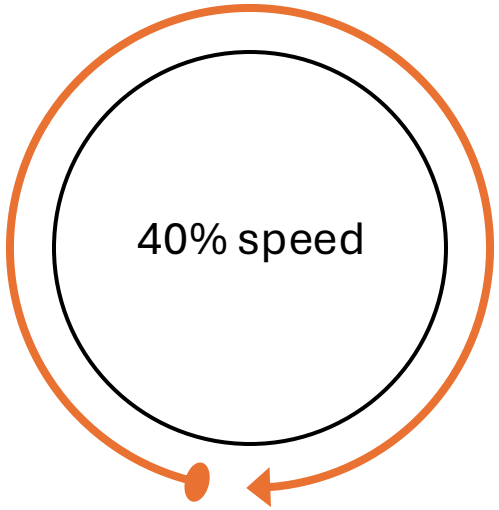
Swim Stroke



Power/return Stroke

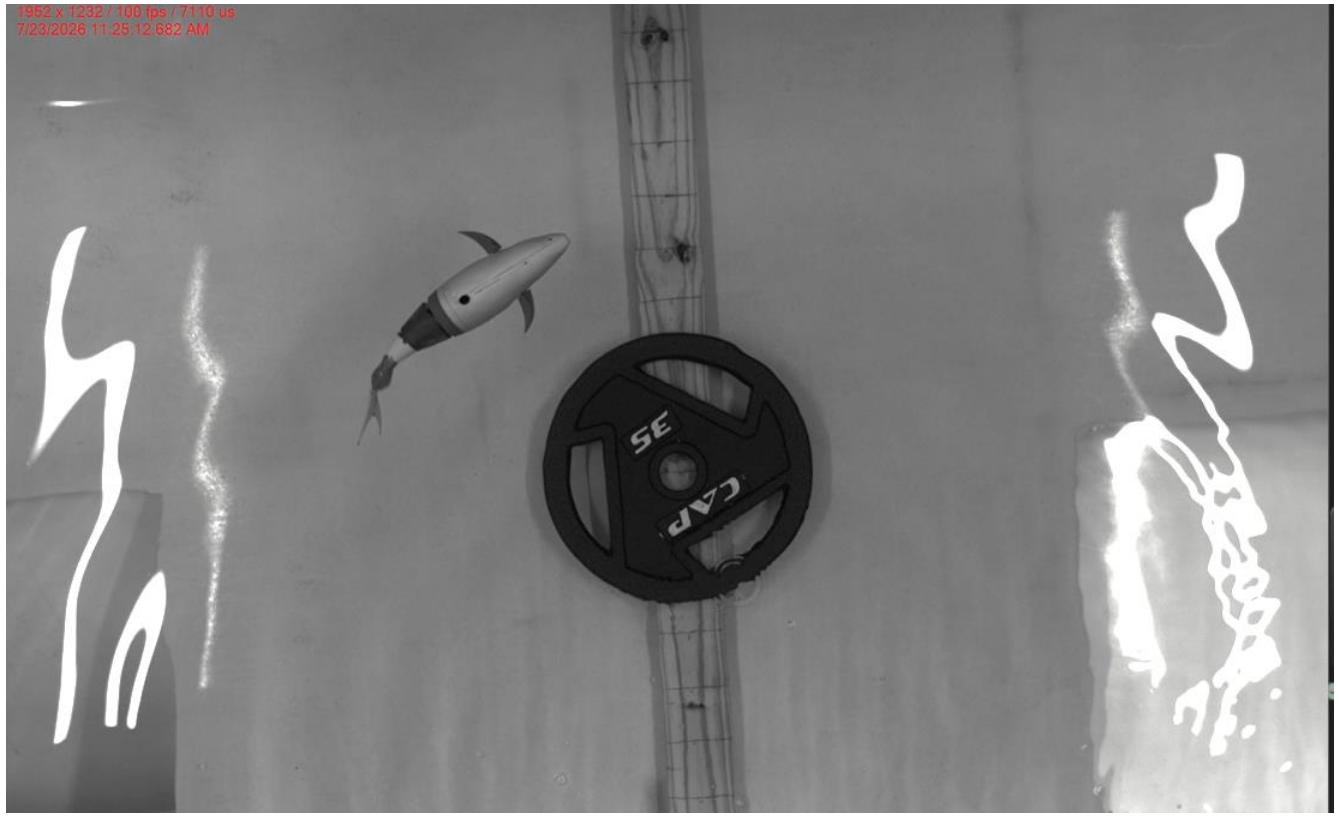


Swim Stroke



Experimental Setup

- Tunabot vehicle is fully free swimming
- Vehicle begins each test at approximately half the depth of the tank
- High speed camera mounted above the tank





Experimental Setup: Configuration

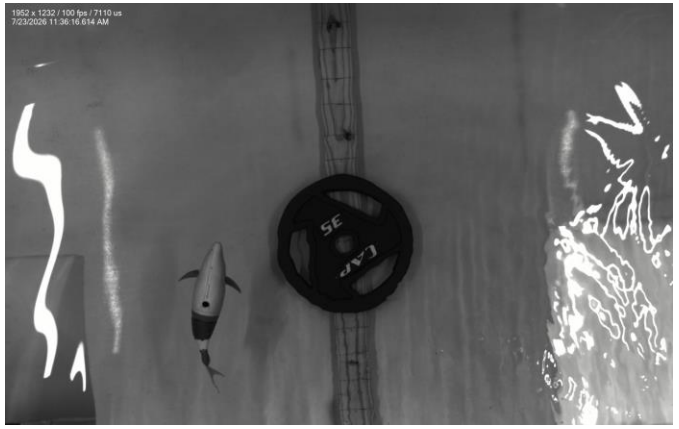
Constants:

- Front angle (where it starts its power stroke): 0 degrees
- Pectoral fin angle: -45 — 45 degrees
- Back angle (where it ends its power stroke): 50 degrees

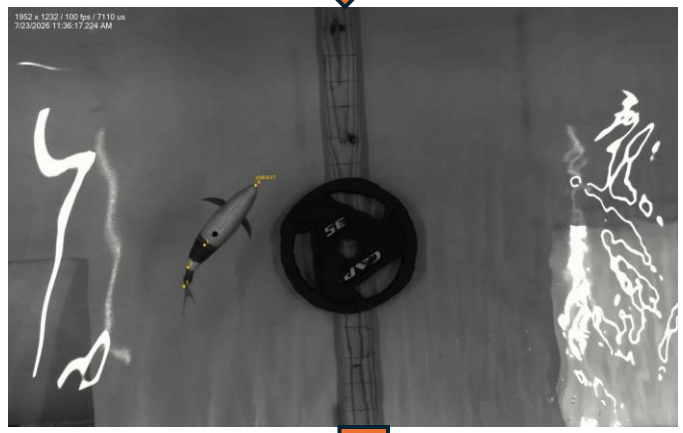
Variable:

- Motor speed: PWM 180 — 255

How data was analyzed and visualized



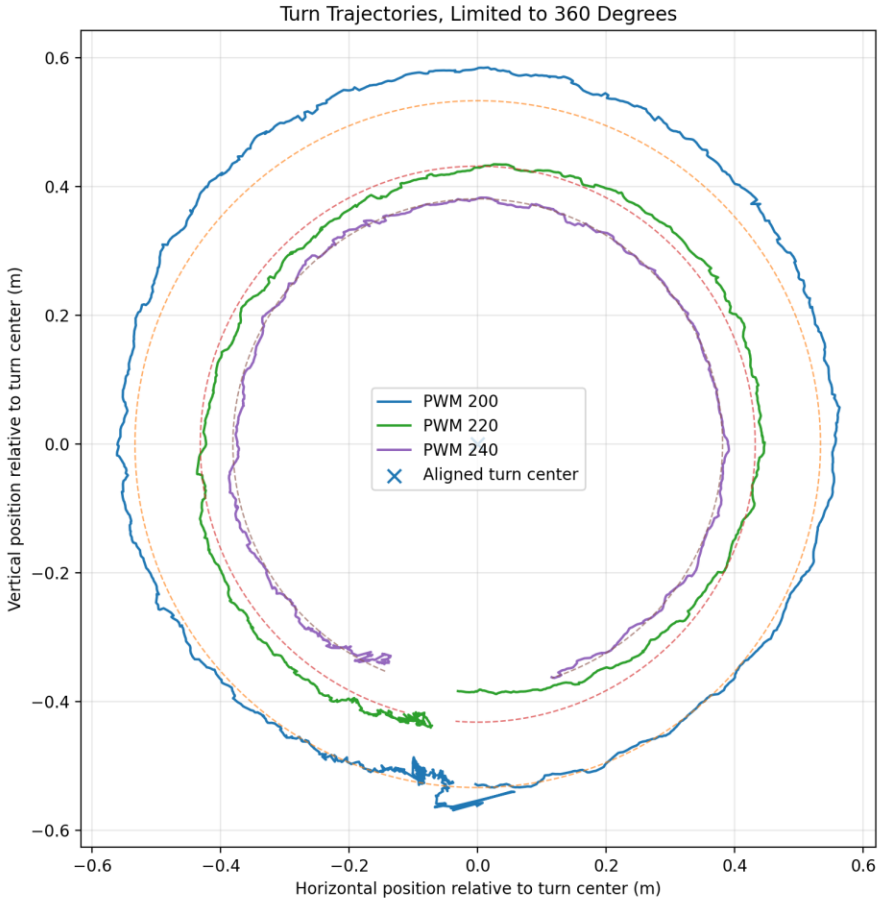
DeepLabCut™:
a software package for
animal pose estimation

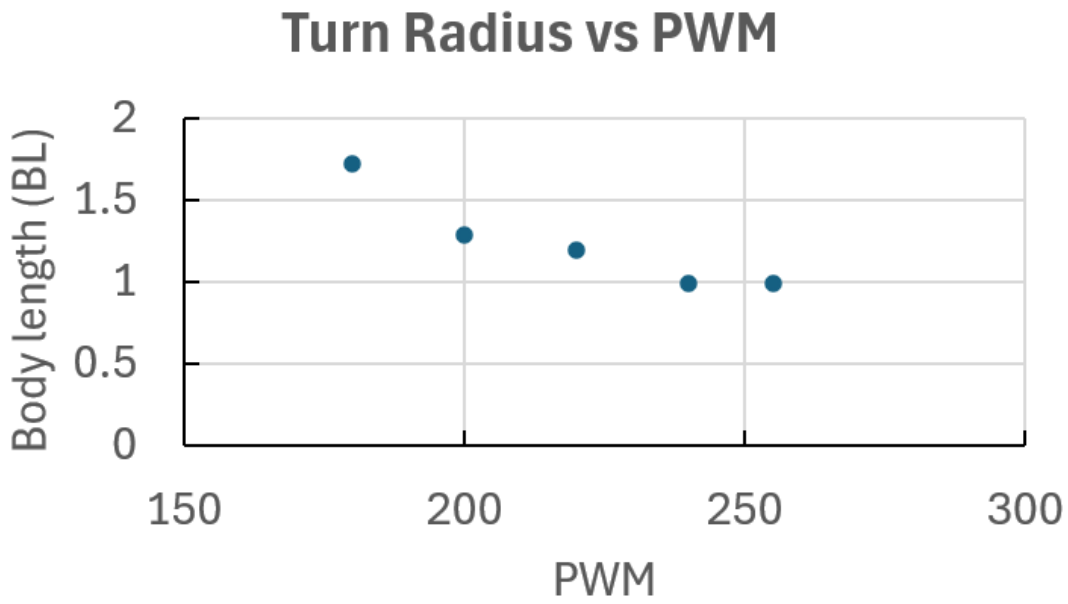


CSV

Rotational Speed,
Tail Frequency,
Rotation per Flap

Quantitative Results: Turn Efficacy by Speed



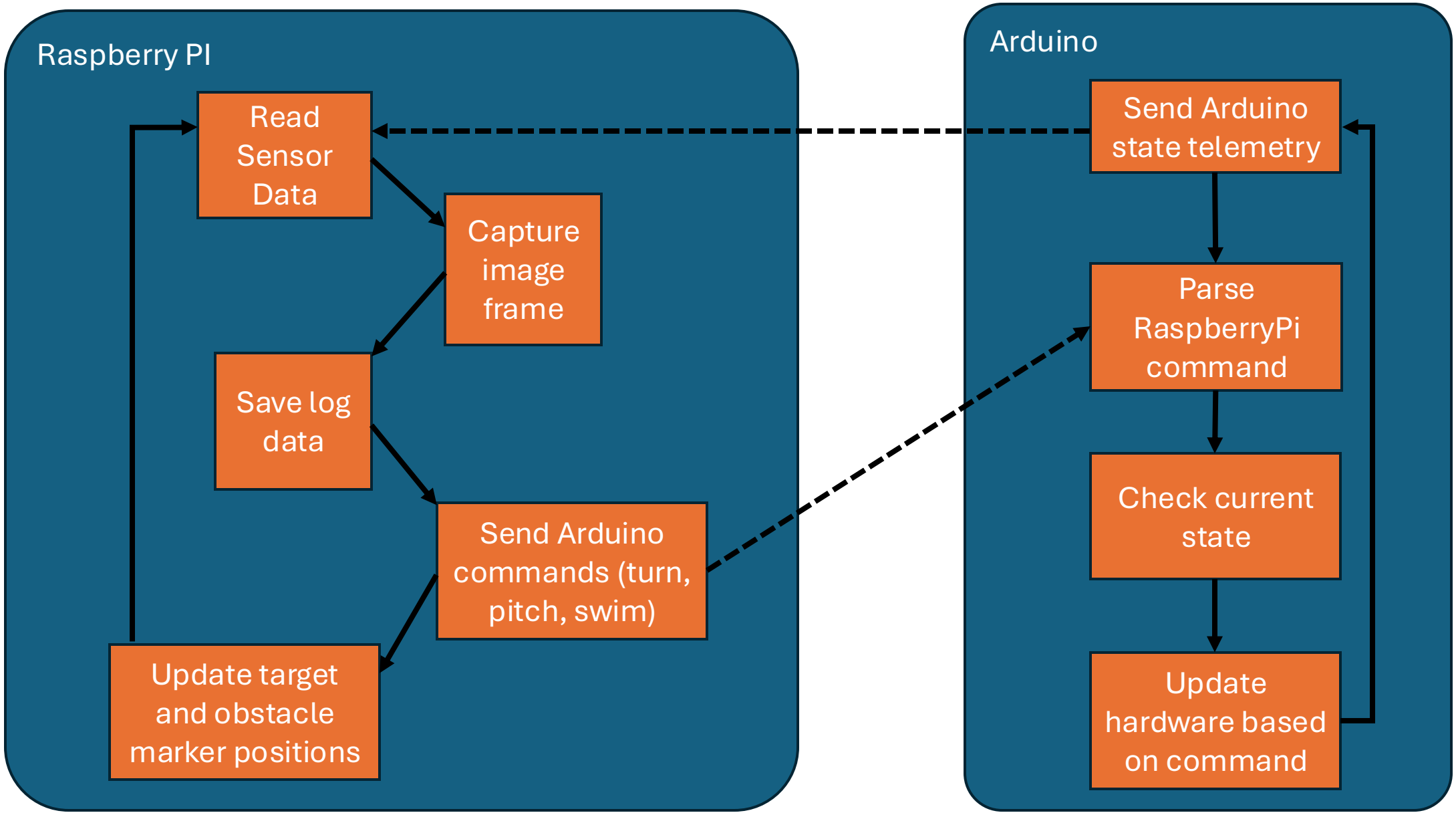


Note: At 180 PWM, vehicle could not hold depth and dragged along floor resulting in an inconsistent turn
Note: At 255 PWM, vehicle turn rates were insignificantly different to PWM 240

Future Improvements

- Use of pectoral fins to act as brakes to decrease turning radius
- Find methods to maintain depth while turning without increasing pitch angle

Practical Utility: Navigation Control Flow




id=2

Results

- 100% success rate over 10 tests with random marker placements

Future Improvements

- Use of AI to recognize diverse objects and improve image processing speed